Fundamental Computer Science (common with M1 Info)

Alastair Abbott

Fundamental Computer Science

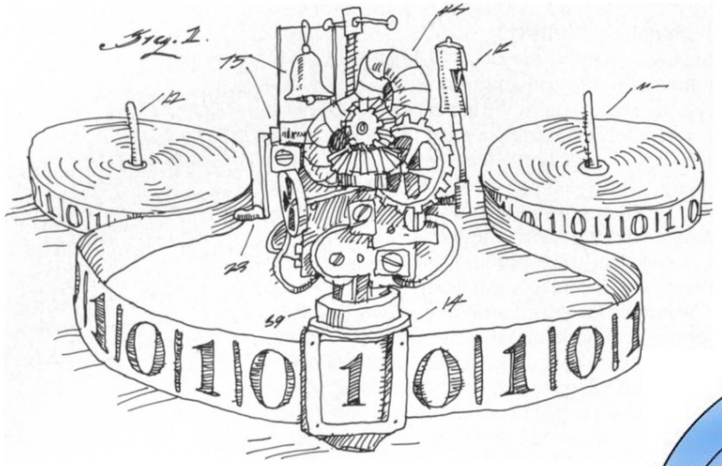
Foundational motivation:

- What does it mean to “compute”? What is an algorithm?
- What problems can be solved by a computer? With what resources?
- Are some problems intrinsically harder than others?

Theoretical computer science to the rescue:

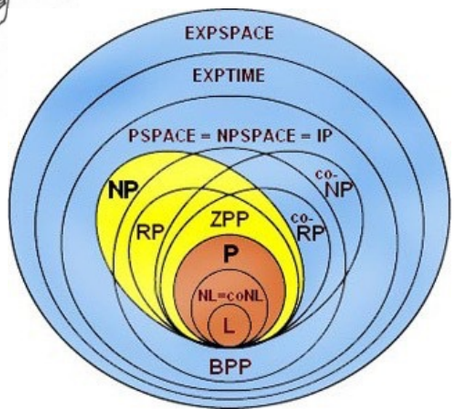
- Abstract away details of specific hardware and languages
- Formal models of computation
- Mathematical formulation & classification of problems

Course content



First half-semester:

- Turing machines: a universal computational model
- Introduction to computational complexity
- Hard problems and NP-completeness



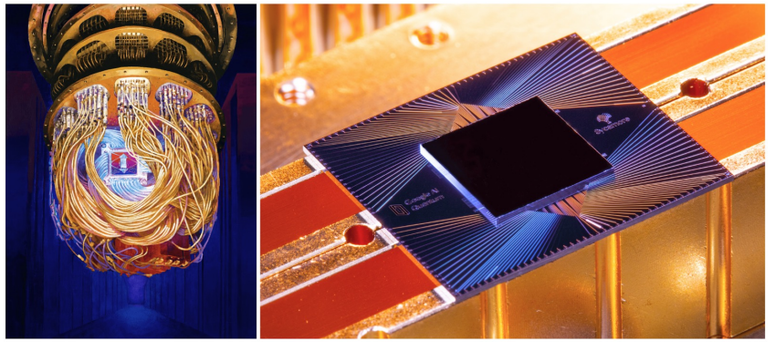
- **Objective:** Formalise and reason mathematically about computation and the difficulty of computational problems; understand how different problems are related

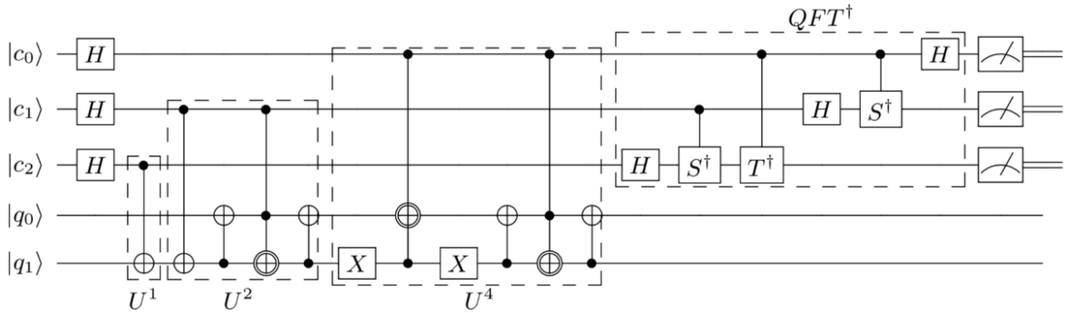
Course content

Second half-semester

Introduction to advanced topics in theoretical computer science

- Approximation algorithms
- Probabilistic algorithms
- Circuit models of computation
- Quantum computing





- **Objectives:** Gain a more nuanced theoretical perspective on computational problems and study some novel ways to approach and analyse them

Course structure

- **Lecturer:** Alastair Abbott
- **Structure:** 11 courses (1.5 hours CM, 1.5 hours TD)
  - In English
  - No labs or programming
- **Evaluation:**
  - 70% exam, 30% internal: 1 written HW & topical group presentations on research articles and more advanced topics

- **Prerequisites:**
  - Mathematical formalisation, reasoning and proofs
  - Basic discrete mathematics (e.g., graphs), logic, linear algebra
  - A curiosity to understand the theoretical foundations of computer science

- Who is it for?
  - Anyone wanting to understand the foundations of computer science and gain new perspectives
  - Continuation in more mathematical aspects of computer science (ORCO, cybersecurity, ...)